

ENGINE ARCHITECTURE / FINANCIAL SERVICES DATA

Purpose-built engines for financial services data.

For a decade, financial institutions have consolidated analytics onto a handful of general-purpose cloud warehouses and lakehouses. These platforms are excellent at what they were built for. But financial services data has quirks of its own: point-in-time correctness, bitemporal history, entity-centric joins, dense rule sets and strict residency constraints. Expressing those quirks on a general-purpose engine is always possible, and it is often expensive. Four open building blocks, the DAVE stack (DataFusion, Arrow, Vortex, Embedded), have changed the economics of building an engine for exactly this data. QBricks applies the approach in EOS, the engine at the heart of the platform.

4

open layers you inherit rather than build: DataFusion, Arrow, Vortex and an Embedded architecture

~70%

faster queries reported by Polar Signals after adopting Vortex for storage

16.5s

for 866M records through the full TPC-H suite on one VM, with EOS, the engine inside QBricks

FINANCIAL SERVICES DATA IS NOT GENERIC DATA

01

Time is not one dimension

Regulators and auditors ask what an institution knew at a given moment, not just what is true now. That means as-of joins, valid-time and transaction-time history, and replaying a report exactly as submitted, even after upstream restatements. General-purpose engines get there through careful modelling, snapshot tables and repeated scans, which multiply both storage and compute.

02

Entities, not rows

Customer, counterparty, beneficial owner and account are resolved across many source systems. Financial crime, KYC and trade-based money laundering analytics are about joining evidence to a resolved entity, often through graph-shaped relationships. These joins are central to the workload, not an edge case.

03

Dense rules and contracts

A mid-sized bank can operate thousands of data quality rules, each evaluated on every load, plus BCBS 239 lineage obligations and, increasingly, formal data contracts such as the Open Data Contract Standard (ODCS). On a general-purpose platform, each rule is frequently another query over another full scan.

04

Skewed, sparse activity

Most customers transact rarely, while a small number of correspondents, merchants and market counterparties generate most of the volume. The analytically interesting questions often touch a scattered subset of rows and columns that no single physical ordering can keep together.

05

Residency and sovereignty

Data protection and localisation rules in the EU, the Kingdom of Saudi Arabia and the UAE, together with DORA's expectations on ICT third-party risk, increasingly require compute to go to the data rather than the reverse. Egress is a cost line and, in some jurisdictions, a compliance problem.

06

The painful middle ground

Historically an analytical database was a very large, costly and slow project, justified only where thousands of organisations shared the need. Financial services sits in the middle ground: each institution's needs are specific, the regulatory workloads are heavy and recurring, and the gap between a general-purpose platform and an expertly optimised one grows material as data scales.

MEET DAVE: BUILD THE PARTS SPECIFIC TO YOUR DATA, INHERIT THE REST

In a September 2026 essay, Tim Poterba, formerly a lead engineer on the Hail genomics platform at the Broad Institute and now building the Phoebe database, argues that a database optimised for a specific domain has shifted from a multi-year undertaking to a reasonable proposition. Each layer of the DAVE stack is designed to be extended at exactly the points where domain optimisation pays off: query plans and joins, table layouts, compute kernels, storage encodings and distributed architecture. His own population genomics database is the production evidence: trillions of sparse, skewed records, with a representative search returning in a couple of seconds from a stateless container reading about 1 GB across roughly 200 files in S3.

WHAT YOU INHERIT	WHERE YOU EXTEND IT
D DataFusion The query engine An embeddable, high-performance SQL query engine: planner, optimiser and vectorised execution, competitive with the leading engines on the ClickBench benchmark.	Custom logical and physical plans, table providers, specialised operators and joins: native as-of joins, and plans that reuse resolved-entity keys rather than recomputing matches.
A Arrow The in-memory format The de facto columnar in-memory format and compute kernel library, approaching its tenth anniversary, with zero-copy interoperability across the analytics industry.	Domain-specific compute kernels written against a stable format: identifier and reference-data checks, and thousands of quality rules evaluated in a single pass.
V Vortex The storage engine A columnar file format and storage engine with modern encodings and built-in object-store I/O, created by SpiralDB and donated to the Apache Software Foundation.	Custom encodings, layouts and filter pushdown that skip data before it is materialised, so scoped queries across wide, sparse tables never decode what they do not need.
E Embedded The deployment model Other people's distributed systems: containers, serverless, batch schedulers and managed clouds, with their security, identity and operational controls.	Thin wrappers that run the same engine wherever the workload and the data are: on premise, in a sovereign cloud region or on an analyst workstation.

THE ECONOMICS

Less compute per question

Pushing selections into storage, evaluating rules in single passes and maintaining views incrementally means reading and processing a fraction of the data a general-purpose plan would touch.

Less data movement

Running compute next to the data reduces egress charges, duplicate copies and the pipelines that maintain them, and simplifies residency compliance.

Less engineering to build and maintain

Because DAVE supplies the query engine, memory model, storage engine and I/O layer, a small team with domain and database expertise can deliver and sustain what previously needed a large platform organisation.

How QBricks applies the approach: EOS

QBricks is an agentic metadata management and data quality platform for financial services. EOS is the engine inside it, and it is built on the DAVE stack. We did not set out to build a general-purpose database. We inherited query execution, in-memory compute and storage from DataFusion, Arrow and Vortex, and concentrated our engineering on what is specific to financial services: data contracts, quality rules, point-in-time correctness, incremental materialised views and the metadata that ties them to governance evidence.

EOS is embedded wherever an institution needs it, on premise, in a sovereign cloud region or on an analyst's workstation, so processing happens where the data already resides. QBricks interfaces with Databricks, Microsoft Fabric, Snowflake and on-premise platforms, including through SQL push-down, and complements rather than replaces them. Entity resolution for financial crime use cases is delivered through a productised Quantexa integration. A layer-by-layer explanation of how EOS extends each part of the stack is at qbricks.ai/eos.

WHAT DAVE DOES NOT DO FOR YOU

A team that knows both

DAVE is not a shortcut to a finished product. Realising the benefit means understanding the domain and database internals: how to turn a business question into a precise query, how data should be partitioned and encoded, which joins should run first. Poterba's own engine is around 70,000 lines of Rust above the DAVE layers.

Supportability

Open-source foundations reduce build effort but not the need for a vendor accountable for the engine built on top of them, with defined support, security patching and release management.

Third-party risk

Open-source components still belong in the ICT third-party and software bill-of-materials records that DORA and local regulators expect. Apache Software Foundation governance helps, but it does not replace an institution's own assessment.

Governance above the engine

Speed is only valuable if results are trusted. Lineage, data contracts, quality evidence and audit trails have to be first-class features, not afterthoughts.

You no longer have to build a whole database to get a database built for your domain.

For data leaders, the practical question is no longer whether a purpose-built engine is affordable. It is which workloads to move first. QBricks 5x5 Ignite is a no-cost engagement that answers that question on five of your own datasets and five of your own rules.

Request a demo →

REFERENCES

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